

Claims Development and Reserving

Chain-Ladder and Random Forest Approaches

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Data Science for Non-Life Insurance — HEC Lausanne, Spring 2026
Group D

31 May 2026

Abstract

We estimate outstanding claim reserves for a portfolio of 33,977 non-life insurance claims from the CAS Loss Reserve Database (accident years 2009–2023) using both chain-ladder and random forest approaches. The standard non-adjusted chain-ladder yields a reserve estimate of **USD 265 M**, while the inflation-adjusted version gives **USD 261 M**. Diagnostic analysis and back-testing reveal systematic calendar-year effects and persistent underestimation linked to residual claim super-inflation. To incorporate claim-level heterogeneity, we develop a three-part random forest model combining payment probability, severity, and closure dynamics. After correcting for pure IBNR claims, the reserve estimate increases from **USD 162 M** to **USD 342 M**. A variance-based credibility combination of the inflation-adjusted chain-ladder and the RF+IBNR model produces a blended estimate of **USD 291 M**, with corrected reserves broadly lying in the range **USD 280–340 M**.

1 Introduction

A central challenge in non-life insurance is to estimate the future cost of claims that have already occurred but are not yet fully resolved. This outstanding claim reserve includes both reported but not settled claims (**RBNS**) and claims not yet reported (**IBNR**).

One of the standard actuarial approaches is the chain-ladder method, which projects future payments using aggregated development patterns. While simple and widely used, it ignores claim-level information and relies on stability assumptions that may fail in the presence of inflation or structural change.

In this project, we complement the chain-ladder with an individual-claim approach based on random forests, and compare the two perspectives. We adopt a structured approach: we start from the data, identify model limitations through diagnostics, apply targeted corrections, validate the results, and finally compare and combine the methods. Our focus is on obtaining reliable reserve estimates in the presence of inflation, censoring, and incomplete claim information.

2 Data and Exploratory Analysis

2.1 Dataset description

The dataset is drawn from the CAS Loss Reserve Database and contains 84,252 (claim, development year) records for 33,977 unique claims. Claims span accident years 2009 to 2023 and development years 1 to 15.

Key variables include the accident year (**Acc_Yr**), development year (**Dev_Yr**), reporting lag (**Rpt_Yr_Lag**), incremental payment (**Incr_Payment**), accumulated payment (**Acc_Incr_Payment**), an open/closed indicator (**open_ind**), and a cumulative general inflation index (**Accum_Infl_Index_Mid**) adjusted to 2009 price levels. The payment year is computed as $\text{Cal_Yr} = \text{Acc_Yr} + \text{Dev_Yr} - 1$, which is essential for detecting inflation effects running along calendar-year diagonals.

The raw dataset contains no missing values and the calendar-year identity holds exactly for every row; no records were removed or modified.

2.2 Triangle structure and reporting delays

Figure 1a visualises the aggregated run-off triangle as a heatmap on a logarithmic colour scale. Payments are small at Dev 1, rise steeply through Dev 2–5 (the main settlement period), and become small and sporadic beyond Dev 10. The triangular shape reflects right-censoring: for recent accident years, only the leftmost development columns are observed. AY 2023, for instance, has only Dev 1. Because recent cohorts are the least mature, they require the largest extrapolations from the fewest observations, making them the most uncertain part of the reserve.

The reporting-delay distribution (Figure 1b) peaks at lag 2, with roughly 75% of claims reported within three years. However, a non-negligible tail extends to lag 8 and beyond. For recent accident years, any claim with a lag exceeding the number of elapsed years remains unreported at valuation date. These pure IBNR claims are addressed explicitly in Section 4.2.

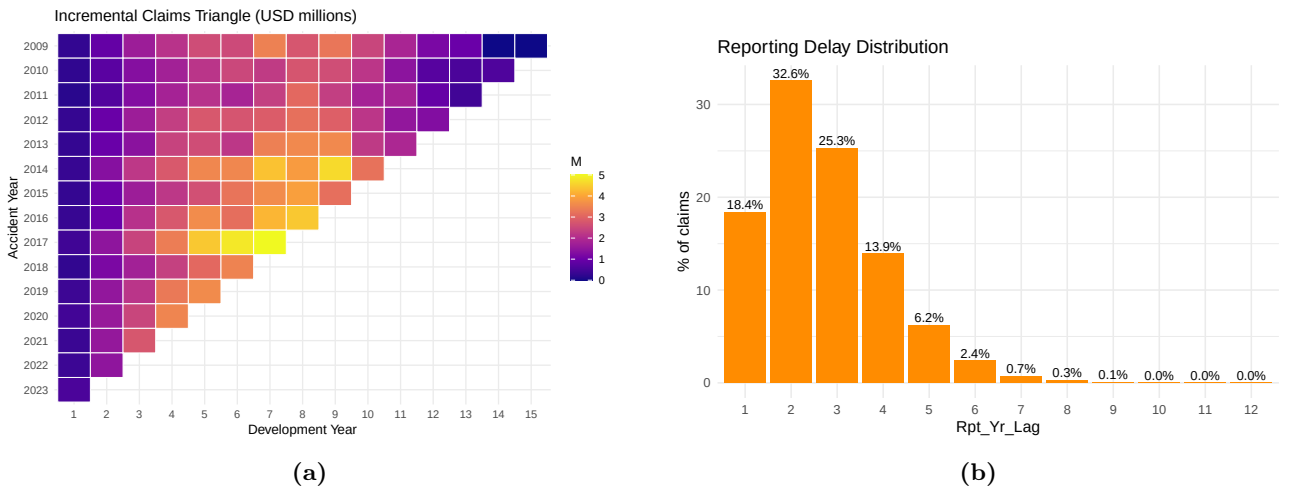


Figure 1

2.3 Claim-size distribution and severity trends

Incremental payments are heavily right-skewed: about 80% of records show zero payment, and positive payments span several orders of magnitude (Figure 2). The long right tail is economically significant; the top 1% of payments drives a disproportionately large fraction of total dollar volume, which motivates the log-transform used in the random forest severity model (Section 4).

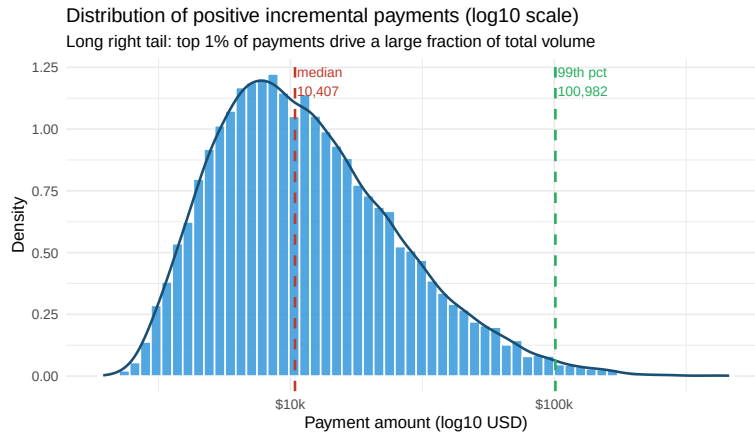


Figure 2

3 Chain-Ladder Analysis and Inflation

3.1 Inflation-adjusted Chain-Ladder and Diagnostics

The inflation-adjusted chain-ladder method yields a reserve estimate of **USD 261 M**. For comparison, the total inflation-unadjusted reserve is USD 265 M.

The validity of this estimate is assessed using the Mack (1993) framework, with all diagnostics performed on the inflation-adjusted (deflated) triangle.

Assumption (A1), independence between accident years, is evaluated using Mack’s rank-correlation test between cumulative payments and link ratios. The resulting p-values are consistently high, providing no evidence against the independence assumption.

Assumption (A2), stability of development factors over time, is assessed through development-year residual patterns. The adjusted triangle shows reasonably stable development behaviour overall, although some variability is visible in early development periods, indicating a mild deviation from full stability.

Assumption (A3), variance proportional to the mean, is assessed by examining the dispersion of the standardised residuals against fitted values (Figure 3). The residuals appear roughly homoscedastic, with no clear evidence of increasing dispersion for larger fitted values. While minor deviations cannot be ruled out, the proportional variance assumption appears broadly reasonable.

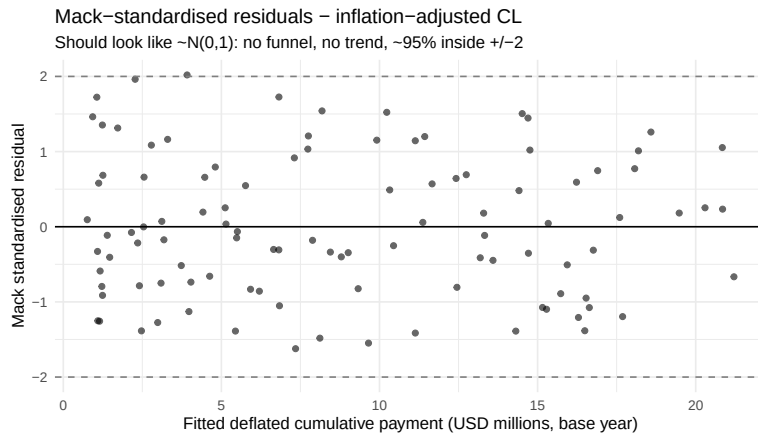


Figure 3

Overall, the diagnostics indicate a reasonable fit of the model assumptions, although minor deviations remain.

3.2 Evidence of claim super-inflation

Even after deflating payments using the general inflation index, claim costs exhibit a clear upward trend over calendar years (Figure 4). A log-linear regression on median deflated payments yields an estimated growth rate of +7.8% per year. This residual increase reflects calendar-year effects not captured by the inflation index alone. As highlighted in the literature, such effects should not be interpreted as pure claims inflation; they may also reflect changes in underwriting practices, reserving behaviour, or claim frequency (Taylor, 2000). We therefore interpret the estimated +7.8% as a composite calendar-year effect.

The standard chain-ladder confounds two distinct effects: genuine claim development (settlement dynamics along development years) and calendar-year price increases (inflation along diagonals). In practice, deflated development factors tend to be systematically lower than nominal ones, suggesting that part of what the nominal chain-ladder attributes to “development” is actually inflation.

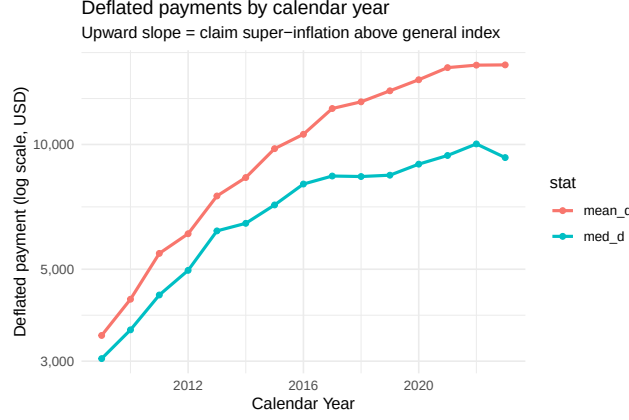


Figure 4: Even after accounting for general inflation, costs grow at +7.8% annually, suggesting the presence of super-inflation above overall price levels.

3.3 Super-inflation scenarios

The inflation-adjusted chain-ladder does not capture this residual calendar-year trend. To assess its impact, we consider three scenarios with additional growth rates g applied to future payments (using the discrete rate $g_{\text{discrete}} = e^{0.0781} - 1 \approx 8.12\%$). The three scenarios yield: Base ($g = 0\%$): USD 261 M; Moderate ($g \approx 4.06\%/yr$): **USD 315 M**; Full ($g \approx 8.12\%/yr$): USD 382 M.

3.4 Retrospective back-test

To quantify the impact of model misspecification empirically, we perform a retrospective back-test across multiple cutoff years (2016–2020). For each cutoff, we fit the chain-ladder on historical data and compare projections to actual payments observed in the subsequent window up to 2023. As the cutoff moves forward (i.e., as more training data becomes available and the prediction horizon shrinks), the underestimation decreases. This pattern confirms that the bias is structural and driven by the calendar-year trend rather than by noise.

4 Individual-Claim Reserving with Random Forests

4.1 Three-part model

The chain-ladder aggregates all claims within an accident year, ignoring claim-level heterogeneity. In contrast, the individual-claim approach differentiates between claims with different characteristics such as reporting delay, payment history, and claim status. We model future payments using a three-part random forest (Breiman, 2001) framework reflecting the classical frequency–severity decomposition:

1. **Payment probability** ($\hat{p}_{\text{pay}}$): a classifier estimating whether an open claim generates a payment in a given development year,

$$\hat{p}_{\text{pay}} = P(S_{i,k} > 0 \mid \text{open at start of } k),$$

trained with balanced class weights to handle the approximately 80% zero-payment rate.

2. **Severity** ($\hat{\mu}_{\text{sev}}$): the expected log-deflated payment conditional on a payment occurring,

$$\hat{\mu}_{\text{sev}} = E[\log(S_{i,k}/\text{Idx}_{\text{Cal}}) \mid S_{i,k} > 0],$$

with retransformation to the original scale using the parametric lognormal correction.

3. **Closure probability** ($\hat{p}_{\text{close}}$): a classifier estimating whether a claim closes during a given development year,

$$\hat{p}_{\text{close}} = P(\text{closed at end of } k \mid \text{open at start of } k),$$

which determines the effective lifetime of each claim and avoids the overestimation inherent in a naive two-part model.

All models use the same feature set: accident year, development year, reporting lag, a binary indicator for prior payments, the log of accumulated paid amount, and a development ratio. For each claim and each future development year, the expected incremental payment is:

$$\hat{S}_{i,k} = \hat{p}_{\text{open}} \cdot \hat{p}_{\text{pay}} \cdot \exp(\hat{\mu}_{\text{sev}} + \hat{\sigma}^2/2) \cdot I_{\text{infl}},$$

where $\hat{p}_{\text{open}}$ tracks the probability that the claim is still open, updated recursively using the closure model. The initial RF model, restricted to reported claims (RBNS), produces an estimated reserve of **USD 162 M**, significantly below the chain-ladder.

4.2 IBNR correction

A random forest trained on reported claims completely misses pure IBNR: claims that have occurred but have not yet been reported. Figure 5a illustrates this gap: when comparing the inflation-adjusted chain-ladder reserve to the RF reserve without IBNR, the RF tracks the chain-ladder closely for mature accident years but collapses for young ones, where a large fraction of claims is still unreported.

We estimate the number of unreported claims for each accident year using the empirical reporting-lag distribution, computed on fully mature accident years (≤ 2013) to avoid truncation bias:

$$N_{\text{IBNR}}(i) = N_{\text{obs}}(i) \cdot \frac{1 - P(\text{Lag} \leq t_i)}{P(\text{Lag} \leq t_i)},$$

where t_i is the number of elapsed years for accident year i . Synthetic IBNR claims are generated by sampling from the conditional lag distribution and projected using the same three-part RF. This adds USD 180 M, increasing the reserve from USD 162 M to **USD 342 M**.

Figure 5b shows the resulting reserve decomposition. For the youngest accident years, pure IBNR accounts for the majority of the reserve. After adding the IBNR component, the RF reserve aligns much more closely with the chain-ladder, confirming that the initial gap was driven by missing unreported claims.

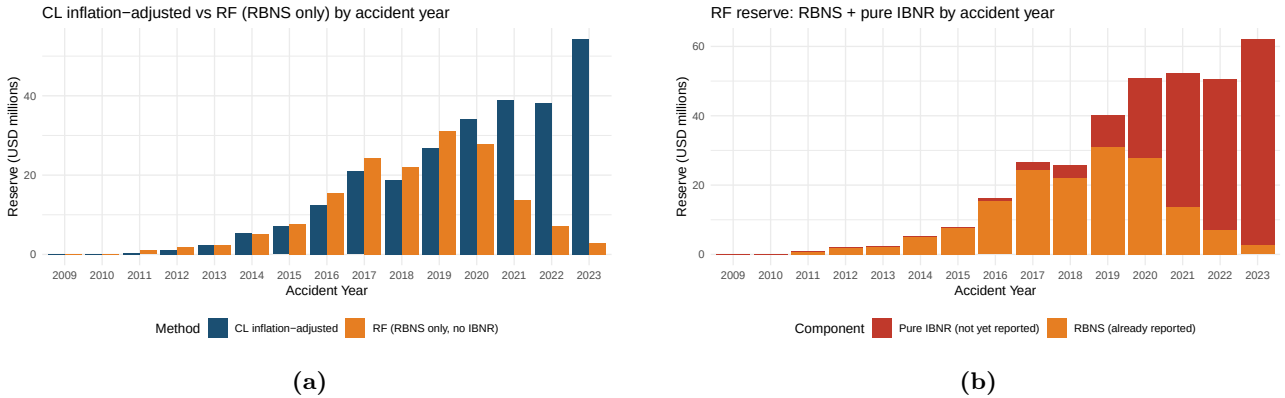


Figure 5

4.3 Right-censoring correction with Cox PH

For recent claims, only a limited number of development years is observed. A claim still open at an early stage is not necessarily long-lasting; it may simply not have been observed long enough. A standard RF closure model, which ignores this right-censoring, tends to overestimate survival probabilities and consequently underestimate future payments.

To address this, we replace the RF closure model with a Cox proportional hazards model (Klein & Moeschberger, 2003), which explicitly accounts for censored observations. The hazard of closure for claim i at time t is

$$h_i(t) = h_0(t) \exp(\beta^\top \mathbf{x}_i),$$

where $h_0(t)$ is an unspecified baseline hazard, $\mathbf{x}_i$ the covariates, and β estimated via partial likelihood. The covariates include accident year, reporting lag, payment indicator, and log accumulated payment.

The corresponding survival function is

$$S_i(t) = [S_0(t)]^{\exp(\beta^\top \mathbf{x}_i)},$$

with $S_0(t)$ obtained via the Breslow estimator. The conditional closure probability is then given by $\hat{p}_{\text{close}}^{\text{Cox}}(d) = 1 - \frac{S_i(d)}{S_i(d-1)}$, which replaces the RF closure probability in the projection of RBNS claims.

The Cox correction is applied only to RBNS claims, as synthetic IBNR claims have no censoring history. Combined with the IBNR adjustment, the Cox-corrected model yields **USD 384 M**, which we interpret as a conservative upper bound due to potential overestimation of long claim durations.

4.4 RF validation

To validate the random forest out of sample, we perform a calendar-year back-test analogous to the chain-ladder procedure described in Section 3.4. For each cutoff year (2016–2020), the RF models are retrained using data up to the cutoff and applied to project future payments for RBNS claims. Predictions are then compared to observed payments over the subsequent period.

Figure 6 compares the total bias of the inflation-adjusted chain-ladder (CL) and the random forest (RF) by cutoff year. The two methods exhibit opposite bias patterns. The CL systematically underestimates reserves at all cutoffs, with errors decreasing as the training sample grows (from -40.6% in 2016 to -3.1% in 2020). In contrast, the RF consistently overestimates, and its bias increases markedly as additional data are incorporated (from $+24.4\%$ in 2016 to $+66.5\%$ in 2020).

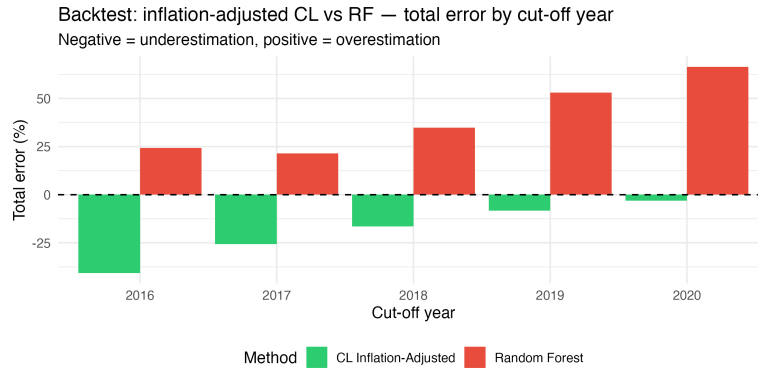


Figure 6: Out-of-sample back-test: total error (%) by cutoff year for inflation-adjusted chain-ladder and random forest models. Negative values indicate underestimation, while positive values indicate overestimation.

5 Combination, and Uncertainty

5.1 Confidence intervals

For the chain-ladder, prediction uncertainty is estimated using the over-dispersed Poisson bootstrap of England & Verrall (1999). Pearson residuals are resampled with replacement and added to fitted values to generate pseudo-triangles, on which the model is refitted.

For the random forest, we adopt a cluster bootstrap by resampling claim IDs with replacement, preserving within-claim dependence. Uncertainty in the number of IBNR claims is incorporated via

$$N_{\text{IBNR}}(i) \sim \text{Poisson}(\lambda_i).$$

At each of the $B = 100$ iterations, the RF models are retrained, the portfolio is projected, and the simulated IBNR component is added. This yields a 95% confidence interval of [\$266 M, \$342 M] around the point estimate of USD 297 M. The distribution is right-skewed, reflecting heavy-tailed severity.

Both procedures capture sampling uncertainty under a stable data-generating process, but do not account for structural changes. Back-testing results (Sections 3.4 and 4.4) suggest that such effects are likely more important in practice. The same per-AY bootstrap outputs feed directly into the variance-based credibility weights of Section 5.2.

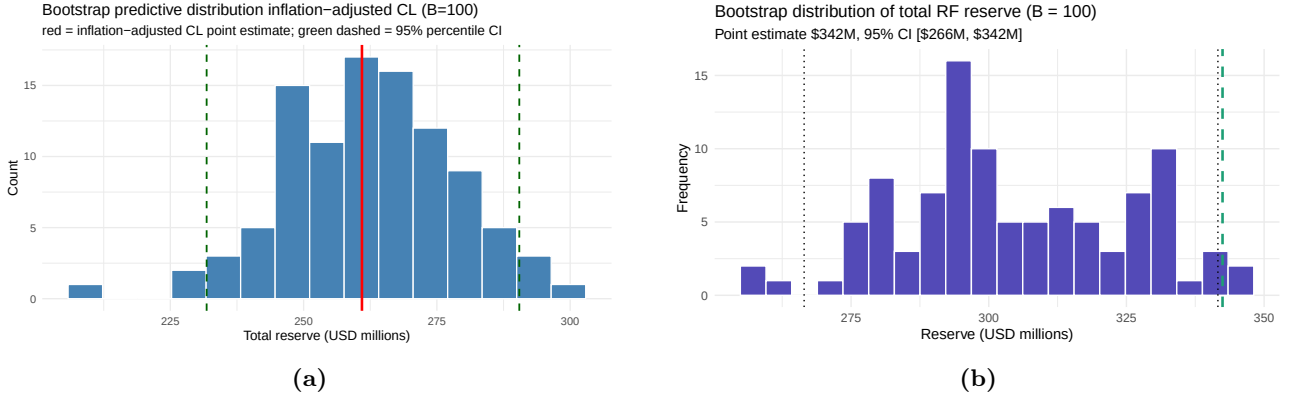


Figure 7: (a) Inflation-adjusted chain-ladder bootstrap, CI [235 M, 290 M] around point estimate USD 261 M. (b) Random forest: the green dashed line marks the RF point estimate; dotted lines mark the 95% CI [266 M, 342 M]. The RF distribution is right-skewed, reflecting heavy-tailed severity.

5.2 Variance-based credibility combination of Chain-Ladder and Random Forest

Backtesting shows that the two methods tend to produce deviations in opposite directions, highlighting their complementarity. In addition, bootstrap results indicate that the random forest produces confidence intervals at higher levels than the chain-ladder, consistent with its higher reserve estimates. Taken together, these findings suggest that the two methods capture different aspects of the underlying risk and that neither dominates across all accident years. We therefore combine them using a credibility weight that varies by accident year. Rather than imposing a fixed maturity-based prior on which method should dominate, we adopt a data-driven approach: for each accident year, we measure how dispersed each method’s reserve estimate is across bootstrap replicates and assign more weight to the method that exhibits lower variability.

Concretely, let $\widehat{\text{Var}}(\hat{R}_i^{\text{CL}})$ and $\widehat{\text{Var}}(\hat{R}_i^{\text{RF}})$ denote the empirical variances of the per-AY reserves across the $B = 100$ bootstrap replicates of Section 5.1. The weight assigned to the random forest for accident year i is

$$z_i = \frac{\widehat{\text{Var}}(\hat{R}_i^{\text{CL}})}{\widehat{\text{Var}}(\hat{R}_i^{\text{CL}}) + \widehat{\text{Var}}(\hat{R}_i^{\text{RF}})},$$

$$\hat{R}_i^{\text{blend}} = (1 - z_i) \hat{R}_i^{\text{CL,adj}} + z_i \hat{R}_i^{\text{RF+IBNR}}.$$

When the chain-ladder is unstable for a given cohort (high bootstrap variance), z_i increases and the blend leans toward the random forest; conversely, when the random forest is more dispersed, z_i decreases and the blend leans toward the chain-ladder. No assumption is made a priori about which method should dominate for which accident years: the bootstrap dispersion decides.

The resulting total reserve is **USD 291 M**, lying within the range defined by the CL inflation-adjusted estimate (USD 261 M) and the RF + IBNR estimate (USD 342 M).

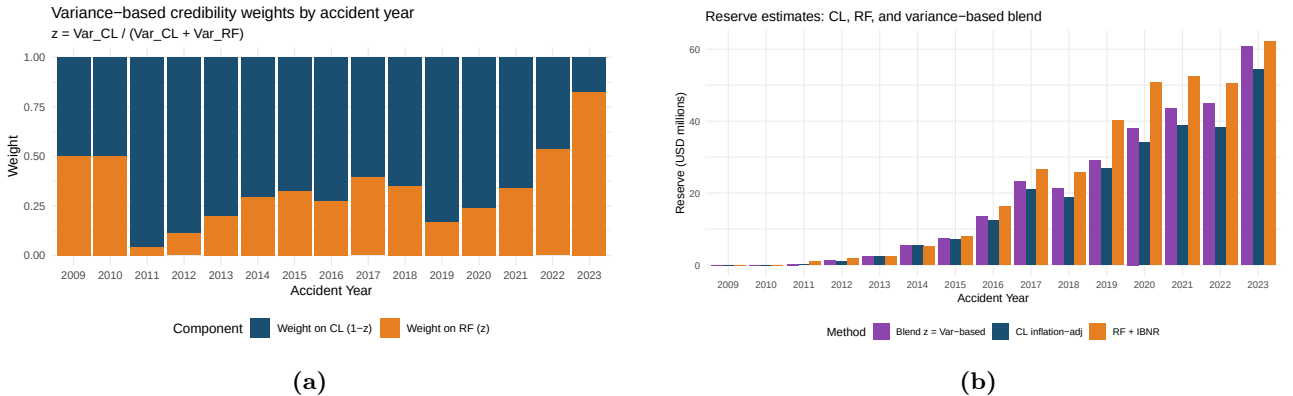


Figure 8: (a) Variance-based weights z_i by accident year: more weight on the method with lower bootstrap variance for that cohort. (b) The blend (purple) lies between the two component estimates.

6 Conclusion

Estimating outstanding claim reserves requires balancing simplicity and realism, and no single method achieves both perfectly. This study brings together two fundamentally different approaches and shows that combining them leads to more robust estimates than relying on either in isolation.

Chain-ladder. The chain-ladder is valued for its transparency, ease of implementation, and stable aggregate projections. Its main limitation is structural: it confounds development dynamics with calendar-year effects such as inflation. As a result, it tends to systematically underestimate reserves in non-stationary environments. Inflation-adjusted variants mitigate this issue, but residual super-inflation (+7.8%/yr above the inflation index) remains unaccounted for unless explicit scenarios are applied.

Random forest. The individual-claim random forest captures claim-level heterogeneity through payment, severity, and closure modelling, and partially absorbs calendar-year trends through the use of deflated payments and accident year as a covariate. However, it is sensitive to missing IBNR (which accounts for USD 180 M of the final estimate) and right-censoring in closure dynamics, both of which require explicit corrections to avoid severe underestimation. Remaining limitations include the assumption of feature independence across development years and the reliance on a stationary lag distribution for IBNR projection; relaxing these would be natural extensions.

Combination. Despite fundamentally different modelling assumptions, the corrected estimates converge to a broadly consistent range. The variance-based credibility blend yields **USD 291 M**, the chain-ladder with moderate super-inflation **USD 315 M**, and the RF with full IBNR **USD 342 M**. The RF + Cox correction (USD 384 M) and the full super-inflation scenario (USD 382 M) define a potential upper bound, while the uncorrected RF (USD 162 M) is clearly downward biased. Figure 9 illustrates this alignment across methods.

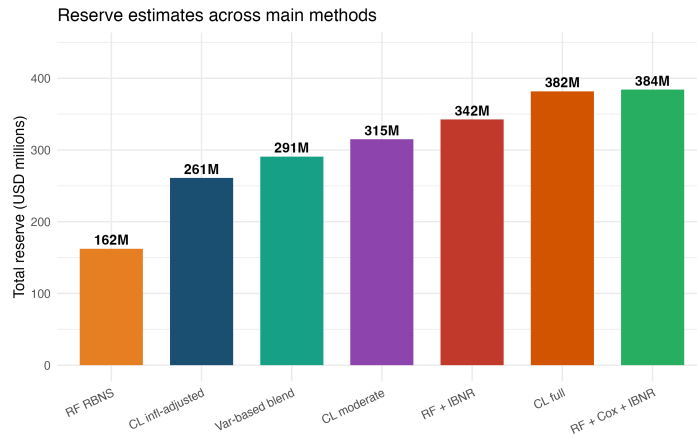


Figure 9

This convergence supports a central estimate in the **USD 280–340 M** range. The variance-based weighting formalises the combination by letting bootstrap dispersion drive the per-cohort allocation: accident years for which one method produces highly variable estimates are automatically down-weighted in favour of the other, without requiring an a priori judgement on which method is more appropriate where.

More generally, these findings extend beyond this dataset: neither method should replace the other, as the chain-ladder provides a transparent and widely accepted actuarial benchmark, while the random forest acts as a diagnostic lens that reveals what aggregate methods miss. Combining the two through variance-based weighting offers a practical and robust reserving framework. Finally, while bootstrap confidence intervals capture sampling uncertainty, they do not account for structural shifts such as inflation regimes, claims handling changes, or regulatory evolution. In practice, such factors are likely to dominate forecast risk, reinforcing the importance of ongoing validation and back-testing. A direction for future work is to extend the dataset, relax some of the present modelling RF assumptions, and formally benchmark our variance-based blending approach with methods like the Bornhuetter-Ferguson.

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